

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
Faculty Name	Dr. Sukanya
Student Name	Kompelli Saanvi
Roll Number	2520030083
Branch	CSE
Course Outcome	6

Case Study Topic: Steel Mill Rod-Cutting Profit Maximization using Dynamic Programming

Word Done Summary:

In this case study, Dynamic Programming (DP) was applied to solve the classic **Rod Cutting Problem** faced by a steel manufacturing company. A 10-foot steel rod must be cut into smaller pieces and sold for maximum profit. Different rod lengths have different selling prices.

Price Table:

Length Price

0	0
1	1
2	5
3	8
4	9
5	10
6	17
7	17
8	20
9	24
10	30

Implementation Details:

1. Maximum obtainable revenue.
2. Optimal cutting sequence.
3. Dynamic Programming table construction.

4. Time and space complexity.
5. Relationship with Unbounded Knapsack.

Result:

Input:

Rod Length = 10

Price = [0,1,5,8,9,10,17,17,20,24,30]

DP Table Generated

dp = [0,1,5,8,10,13,17,18,22,25,30]

Maximum Revenue

30

Optimal Cut Sequence

[10]

Dynamic Programming efficiently computes the maximum obtainable revenue from a steel rod. The DP solution avoids repeated calculations and guarantees the optimal result. For the given price table, selling the entire 10-foot rod without cutting yields the highest profit of 30 units.

Output:

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Maximum Revenue = 30
Optimal Cuts: 10
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GitHub Link:

<https://github.com/saanvikompelli-cmd/DSA-2-Case-Studies>

Student Signature	Faculty Signature
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